

ADAR'S STUDY SERIES

Linear Algebra

Part 5

Linear Independence
Concepts, Tests, and Conceptual Reasoning

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PART 5

Linear Independence

Overview & Goals

- **Goal 1:** Learn what **linearly independent** and **linearly dependent** vectors are conceptually, and be able to check independence computationally.
- **Goal 2:** Use the concept of independence to tackle conceptual examples.

What is Linear Independence?

Definition: Let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \in \mathbb{R}^n$.

The set $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is **linearly independent** if the vector equation:

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_k\vec{v}_k = \vec{0}$$

only has the **trivial solution**:

$$c_1 = c_2 = \dots = c_k = 0$$

In matrix form, $A\vec{x} = \vec{0}$ (where $A = [\vec{v}_1 \ \vec{v}_2 \ \dots \ \vec{v}_k]$ and

$\vec{x} =$

$$\begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}$$

) only has the solution $\vec{x} = \vec{0}$.

Definition: If there exists a solution where **not all** $c_i = 0$, we say the set $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is **linearly dependent**.

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Example 1: Demonstrating Linear Dependence

Given the combination:

$$4 \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix} - 10 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ -1 \\ 13 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Because there exists a non-trivial solution ($c_1 = 4$, $c_2 = -10$, $c_3 = 2$), the set of vectors:

$$\left\{ \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 13 \end{bmatrix} \right\}$$

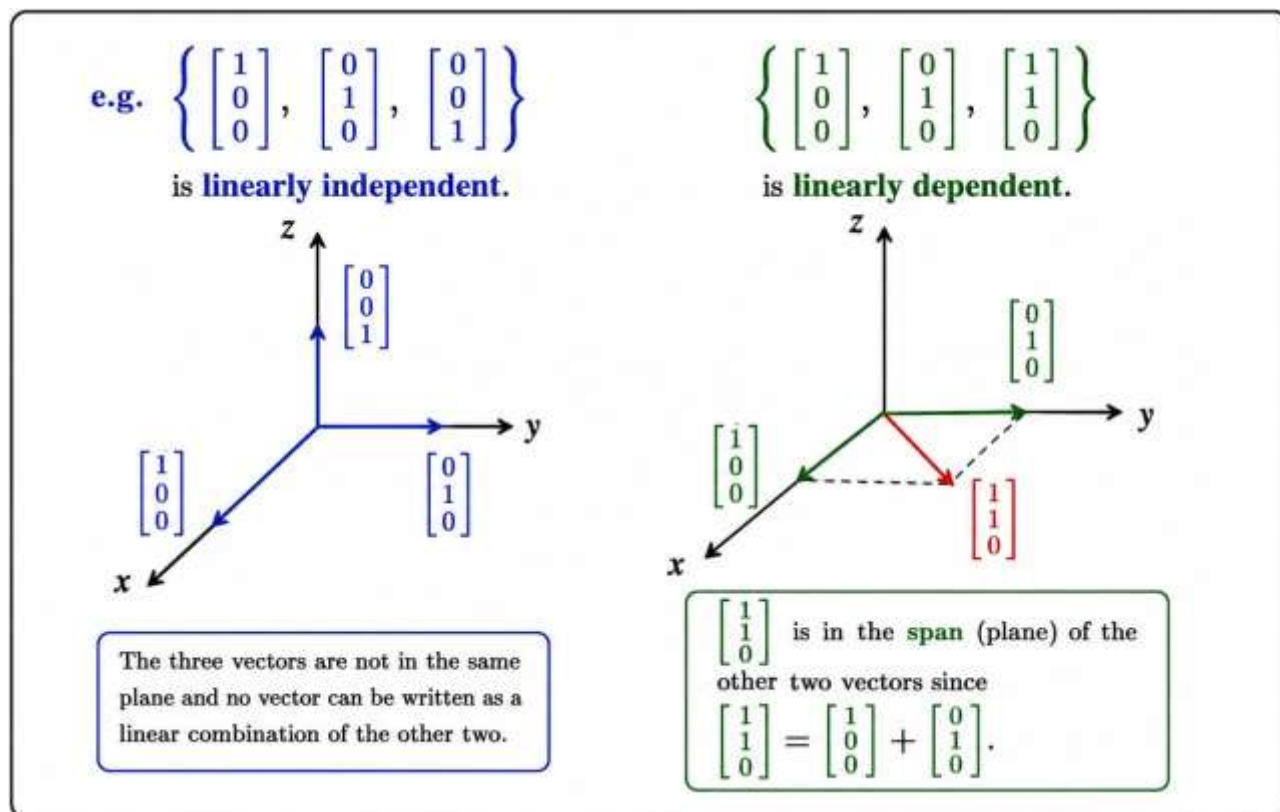
is linearly dependent.

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Intuitive & Geometric Interpretation

Intuitively, $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is linearly independent if the vectors point in "**completely independent**" directions.

Comparison in $\mathbb{R}^3$



Key Theorem & Caution

Theorem: A set $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is **linearly dependent** if and only if **at least one** vector $\vec{v}_i$ is a linear combination of the other vectors.

Caution: It is **not necessarily true** that every vector $\vec{v}_i$ in a dependent set is a linear combination of the other vectors.

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Testing for Linear Independence

Key Criterion:

The vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_k$ are **linearly independent** if and only if $A\vec{x} = \vec{0}$ has a **unique solution** (i.e., $\vec{x} = \vec{0}$).



Matrix A has **no free variables** (i.e., there is a **pivot in every column** of A).

Example 2: Testing Sets for Independence

Problem: Test if the following sets of vectors are linearly independent. If not, express a linear dependence relationship among them.

Part (a)

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \right\}$$

Solution:

Form the matrix A using the vectors as columns and reduce to row-echelon form:

$$\begin{bmatrix} -1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 0 & 4 \end{bmatrix}$$

1. Performing ($R_1 + R_2 \rightarrow R_2$ and $2R_1 + R_3 \rightarrow R_3$):

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & 3 & 3 \\ 0 & 4 & 6 \end{bmatrix}$$

2. Performing ($\frac{1}{3}R_2 \rightarrow R_2$):

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 4 & 6 \end{bmatrix}$$

3. Performing ($-4R_2 + R_3 \rightarrow R_3$):

$$\begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

- **Conclusion:** Since there is a **pivot in every column**, there are no free variables. Therefore, the column vectors are **linearly independent**.

Part (b)

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 12 \\ 3 \\ -4 \end{bmatrix} \right\}$$

Solution:

Form the matrix A and row-reduce to reduced row-echelon form (RREF):

$$\left[\begin{array}{ccc|c} -1 & 2 & 12 & 0 \\ 1 & 1 & 3 & 0 \\ 2 & 0 & -4 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

- **Analysis:** Not all columns have pivots (the 3rd column lacks a pivot), so x_3 is a free variable. Thus, the column vectors are **linearly dependent**.

Finding the Linear Dependence Relationship:

Set up the homogeneous system $AX = \vec{0}$:

$$\begin{aligned} x_1 - 2x_3 &= 0 &\Rightarrow x_1 &= 2x_3 \\ x_2 + 5x_3 &= 0 &\Rightarrow x_2 &= -5x_3 \\ x_3 &= \text{free} \end{aligned}$$

Writing the solution in parametric vector form:

$$\vec{x} = x_3 \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$$

Choosing $x_3 = 1$ gives $x_1 = 2$, $x_2 = -5$, and $x_3 = 1$.

Linear Dependence Relationship:

$$2 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} - 5 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 12 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

PART 5

Conceptual Examples

Example 3

Problem: Can a set of 4 vectors in $\mathbb{R}^3$ be linearly independent? If yes, give an example. If no, explain why.

Answer: No

Proof / Explanation:

Let $\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4 \in \mathbb{R}^3$. Form the matrix A using these vectors as columns:

$$A = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \vec{v}_3 & \vec{v}_4 \end{bmatrix}$$

- **Matrix Dimensions:** Since each vector belongs to $\mathbb{R}^3$, matrix A has 3 rows and 4 columns (3×4).
- **Pivot Constraint:** There are only 3 rows, so the matrix can have at most 3 pivots.
- **Column Analysis:** Since A has 4 columns and at most 3 pivots, **not all columns can contain a pivot** (there is at least one free variable).
- **Conclusion:** Since not every column has a pivot, the column vectors are **linearly dependent**.

Intuition:

There cannot be 4 independent directions in a 3-dimensional space ($\mathbb{R}^3$).

General Theorem:

If a matrix A is $m \times n$ with $n > m$ (i.e., there are more vectors than elements in each vector), then its columns are **always linearly dependent**.

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Example 4

Problem (True / False):

If $\vec{v}_1, \vec{v}_2, \vec{v}_3 \in \mathbb{R}^3$ and $\vec{v}_1$ is **not** a linear combination of $\vec{v}_2$ and $\vec{v}_3$, then $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is linearly independent.

(If true, explain why. If false, give a counterexample.)

Key Theorem:

A set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ is **linearly dependent** $\Leftrightarrow$ **at least one** vector $\vec{v}_i$ is a linear combination of the other vectors.

(Note: We don't necessarily know which specific vector it is.)

Answer: False

Counterexample:

Choose the following vectors in $\mathbb{R}^3$:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

1. Verify the Hypothesis ($\vec{v}_1$ is not a linear combination of $\vec{v}_2$ and $\vec{v}_3$):

Any linear combination of $\vec{v}_2$ and $\vec{v}_3$ gives:

$$a \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Therefore, $\vec{v}_1$ is **not** a linear combination of $\vec{v}_2$ and $\vec{v}_3$.

2. Test for Linear Independence of the Set $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$:

Form the matrix $A = [\vec{v}_1 \ \vec{v}_2 \ \vec{v}_3]$:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- Matrix A has only 1 pivot (in the 1st column).
- It **doesn't** have pivots in each column (Columns 2 and 3 lack pivots).
- Therefore, the set $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is **linearly dependent**.

END OF PART 5 · NEXT: PART 6 — LINEAR TRANSFORMATIONS